

DATA ANALYSIS: survival after HIV

Tijn Jacobs

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We illustrate the proposed framework on a study of antiretroviral therapy for HIV-1 infection, combining randomised data from the AIDS Clinical Trials Group Protocol 175 (ACTG 175) with observational data from the Multicenter AIDS Cohort Study (MACS). HTE estimation is clinically meaningful in this setting because response to antiretroviral therapy varies substantially across patient subgroups, with baseline CD4 count, viral load, and age repeatedly shown to modify treatment effect (Egger et al., 2002). Identifying which patients benefit most or least from a given regimen has direct implications for clinical decision-making, and the underpowering of ACTG 175 for subgroup-level effects makes the case for borrowing from observational data immediate.

ACTG 175 was a randomised trial conducted between 1991 and 1994 comparing four antiretroviral regimens in approximately 2400 adults infected with HIV-1 and CD4 counts between 200 and 500 cells/mm³ (Hammer et al., 1996). The dataset records time-to-event outcomes alongside baseline covariates including age, weight, CD4 count, prior antiretroviral history, and others; it has become a widely used benchmark in causal inference methodology (Tsiatis et al., 2008; Zhang et al., 2012). MACS is a long-running prospective cohort initiated in 1984 that has followed gay and bisexual men across multiple US sites under routine clinical care (Kaslow et al., 1987). Treatment decisions in MACS reflect real-world prescribing rather than randomised allocation and depend on factors only partially captured in the recorded covariates, such as clinician judgment, adherence, and behavioural and social factors, making MACS a canonical example of an HIV cohort subject to unmeasured confounding (Hernán et al., 2000). The complementarity of the two sources is exactly the structure our framework is designed to exploit: ACTG 175 provides clean identification of the heterogeneous treatment effect, MACS contributes sample size and broader covariate support, and the confounding function $c(x)$ absorbs the residual bias arising from MACS's lack of randomisation.

0.1 Cohort and endpoint

ACTG 175 randomised patients to ZDV monotherapy, didanosine (ddI) monotherapy, ZDV plus ddI, or ZDV plus zalcitabine (ddC). We pool the three non-monotherapy arms into a single comparator, because ACTG 175 found ZDV monotherapy inferior to each of them and the three were comparable. This gives a binary contrast between ZDV mono and combination. We construct one analysis cohort by aligning the entry criteria across sources. We restrict the trial to men, matching the all-male cohort, and the cohort to baseline CD4 of 200 to 500 cells/mm³, the trial's entry criterion. We keep only patients who are AIDS-free at baseline, within a common pre-HAART calendar window. This alignment is deliberate. Borrowing across sources is credible only if the two cohorts approximate a common source population, under similar measurement, inclusion criteria, clinical practice, and follow-up (Pocock, 1976). Time zero is randomisation in the trial and the first treated visit in the cohort. The endpoint is event-free survival, the time

Source	<i>n</i>	ZDV mono	Combination	Events	Max. follow-up
ACTG 175 (RCT)	1771	432	1339	447	~3.4 y
MACS (RWD)	373	138	235	227	~25 y
Combined	2144	570	1574	674	

Table 1: The trial-aligned cohort by source and treatment group (ZDV mono versus combination).

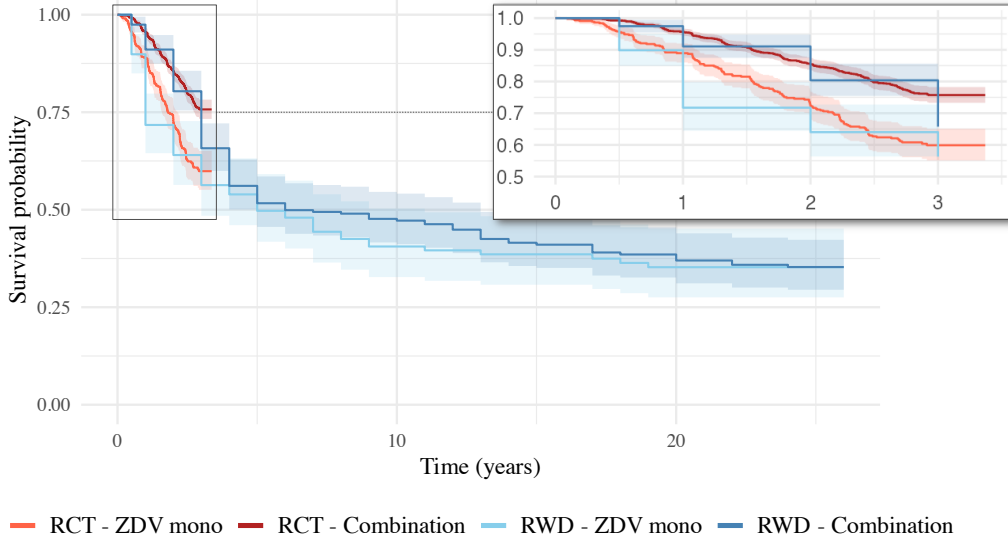


Figure 1: Kaplan–Meier curves for event-free survival, stratified by data source (RCT, RWD) and treatment group (ZDV mono and combination). Main panel shows the full RWD follow-up horizon (~ 25 y); the inset zooms in on the RCT-overlapping window where treatment contrasts are estimated. The RWD curves are approximate because MACS outcomes are interval-censored at the year level (biannual visits, dates redacted in the public release); the fusion analysis accounts for this by passing dead MACS subjects with `interval_censoring_indicator = 1`.

to the first of a 50% CD4 decline, an AIDS-defining event, or death. The trial follows patients for about 3.4 years. The cohort follows them for up to 25 years at annual resolution, so its event times are approximate. We adjust for six baseline covariates: age, CD4, CD8, calendar year, race, and years of prior antiretroviral therapy. Prior therapy is a candidate effect modifier, because ZDV-experienced patients more often carry resistance. Table 1 summarises the two cohorts.

0.2 Survival curves

Figure 1 shows event-free survival by treatment group and source. In the trial, combination separates early and stays above ZDV mono, the established ordering. In the cohort the ordering reverses, and ZDV mono looks favourable at first. This is the signature of confounding by indication, since poorer-prognosis patients were preferentially given combination therapy. The two sources disagree on the raw contrast, and the analysis must reconcile that gap.

Estimand	Trial only	Fusion
Log effect τ	0.56 (0.44 to 0.68)	0.51 (0.36 to 0.67)
Acceleration $\exp\{\tau\}$	1.75 (1.55 to 1.97)	1.67 (1.43 to 1.95)

Table 2: Average treatment effect of combination versus ZDV mono. Posterior mean and 95% credible interval.

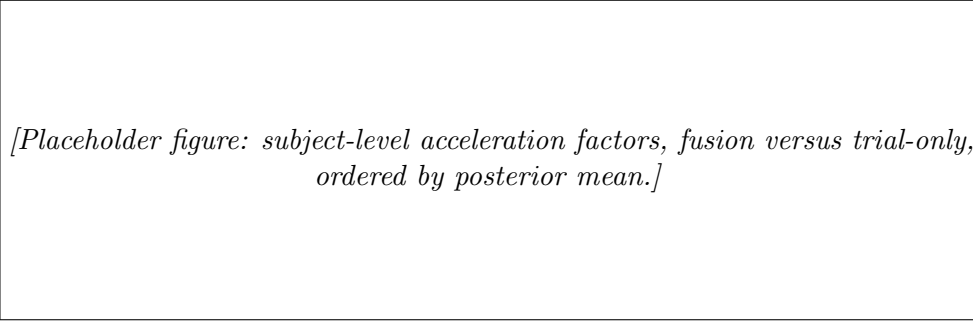


Figure 2: Subject-level acceleration factors, ordered by posterior mean.

0.3 Model and estimands

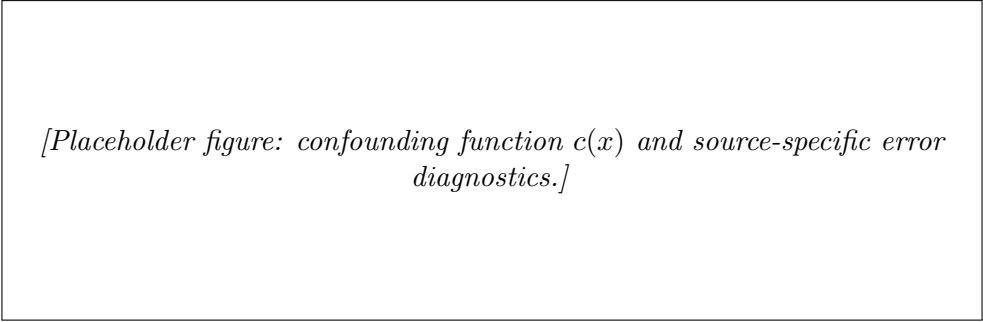
We fit the proposed FusionForest model to the pooled trial and cohort data. We report the conditional average treatment effect $\tau(x)$, the acceleration factor $\exp\{\tau(x)\}$, and the population average treatment effect (ATE). The acceleration factor is the primary summary: it rescales event-free survival time by a constant and reads directly on the clinical scale. As a single-source benchmark we fit a CausalShrinkageForest to the trial alone. We summarise each estimand by its posterior mean and a 95% credible interval. Prior and sampler settings follow the methodology.

0.4 Results

Combination therapy prolongs event-free survival relative to ZDV mono. The fusion estimate of the average acceleration factor is 1.67 (95% CrI 1.43 to 1.95). The trial-only benchmark is close at 1.75 (95% CrI 1.55 to 1.97), so the cohort does not overturn the trial. Table 2 reports both estimands on the log and acceleration scales.

The effect is heterogeneous across patients. Figure 2 orders the subject-level acceleration factors by posterior mean. *[Placeholder: describe the spread and the main effect modifiers, for example the strongest benefit at low baseline CD4 and in treatment-naïve patients.]* *[Placeholder: state whether fusion sharpens the subject-level intervals relative to the trial-only fit.]*

The confounding function isolates the cohort’s bias. *[Placeholder: report the sign and size of $c(x)$ from Figure 3, and confirm it matches the early ZDV-mono advantage in Figure 1.]* The trial and cohort take different residual distributions, which supports the source-specific error. *[Placeholder: summarise the diagnostics, for example the number of active mixture components per source.]*



[Placeholder figure: confounding function $c(x)$ and source-specific error diagnostics.]

Figure 3: Confounding function and source-specific error diagnostics.

0.5 Interpretation

The fusion analysis recovers the trial’s conclusion and extends it. Combination therapy is superior to ZDV mono, and the benefit varies across patients *[placeholder: name the leading modifier]*. The confounding function lets the cohort contribute prognostic information and long-term follow-up without importing its confounding. The result illustrates the framework’s aim: trial-grade identification with the reach and extended follow-up of real-world data.

References

- Egger, M., May, M., Chêne, G., Phillips, A. N., Ledergerber, B., Dabis, F., Costagliola, D., D’Arminio Monforte, A., de Wolf, F., Reiss, P., et al. (2002). Prognosis of HIV-1-infected patients starting highly active antiretroviral therapy: a collaborative analysis of prospective studies. *The Lancet*, 360(9327):119–129.
- Hammer, S. M., Katzenstein, D. A., Hughes, M. D., et al. (1996). A trial comparing nucleoside monotherapy with combination therapy in HIV-infected adults with CD4 cell counts from 200 to 500 per cubic millimeter. *New England Journal of Medicine*, 335(15):1081–1090.
- Hernán, M. A., Brumback, B., and Robins, J. M. (2000). Marginal structural models to estimate the causal effect of zidovudine on the survival of HIV-positive men. *Epidemiology*, 11(5):561–570.
- Kaslow, R. A., Ostrow, D. G., Detels, R., Phair, J. P., Polk, B. F., and Rinaldo, C. R. (1987). The Multicenter AIDS Cohort Study: rationale, organization, and selected characteristics of the participants. *American Journal of Epidemiology*, 126(2):310–318.
- Pocock, S. J. (1976). The combination of randomized and historical controls in clinical trials. *Journal of Chronic Diseases*, 29:175–188.
- Tsiatis, A. A., Davidian, M., Zhang, M., and Lu, X. (2008). Covariate adjustment for two-sample treatment comparisons in randomized clinical trials: a principled yet flexible approach. *Statistics in Medicine*, 27(23):4658–4677.
- Zhang, B., Tsiatis, A. A., Laber, E. B., and Davidian, M. (2012). A robust method for estimating optimal treatment regimes. *Biometrics*, 68(4):1010–1018.

1 Overview

This section is a working scaffold. It records the planned structure of the data-analysis section and the facts to keep fixed while we draft. We write the section one paragraph at a time, then fold this overview in or remove it at the end.

We answer one question. What is the effect of zidovudine (ZDV) monotherapy versus the pooled combination regimens on event-free survival, and how does it vary across patients? We fuse the ACTG 175 trial with the MACS cohort using FusionForest. The trial identifies the treatment effect. The cohort extends follow-up and reflects treatment in routine care. The confounding function shields the effect from the cohort's confounding by indication.

Planned structure

1. **Lead paragraph.** The question and the fusion strategy in a few sentences.
2. **Data and cohort** (Table 1). The two sources, the binary treatment contrast, the EFS endpoint, the trial-aligned cohort, the covariates, and the follow-up horizons.
3. **Survival curves** (Figure 1). Kaplan–Meier by arm and source. The trial shows combination therapy ahead. The cohort shows the reverse, which motivates the confounding function.
4. **Model and estimands.** The four-forest AFT decomposition, the source-specific error, and the reported estimands: CATE $\tau(x)$, the acceleration factor $\exp\{\tau(x)\}$, and the ATE. The RCT-only comparator.
5. **Results.**
 - ATE, fusion versus RCT-only (Table 2, Figure 2).
 - Heterogeneity in $\tau(x)$ (Figure 3).
 - Confounding function, source deviation, and source-specific error (Figure 4).
6. **Interpretation.** Tie the results back to the goals set in the introduction.

Notes to remember

- **Endpoint.** Call it EFS throughout. Do not foreground that the current RWD side is overall survival. The numbers are provisional.
- **Style** (`general/WRITING_GUIDE.MD`). Short active sentences. Topic sentence first. **We** . . . openings. Few signal words. No em-dashes. Report a point estimate with a 95% credible interval, never a *p*-value. `booktabs` tables.
- **Cohort.** RCT (ACTG 175, male) $n = 1771$: $Z=0$ (ZDV mono) 432, $Z=1$ (combination) 1339, with 447 events. RWD (MACS, baseline CD4 200–500) $n = 373$: $Z=0$ 138, $Z=1$ 235, with 227 events. Combined $n = 2144$. RCT horizon ≈ 3.4 y, MACS ≈ 25 y.
- **Covariates.** Age, baseline CD4, baseline CD8, calendar (anchor) year, race, prior antiretroviral years. Prior ARV exposure is the flagged effect modifier.
- **Numbers** (`analysis 1/analysis_summary.txt`). ATE on the log scale: RCT-only 0.559 [0.440, 0.679], fusion 0.508 [0.358, 0.670]. Acceleration factor: RCT-only 1.75 [1.55, 1.97], fusion 1.67 [1.43, 1.95]. $AF > 1$ means ZDV monotherapy is inferior.

- **Figures.** Figure 1 is page 2 of `analysis 0/survival_curves_manuscript.pdf`. Figures 2–4 are pages 1–4 of `analysis 1/analysis_figures_050.pdf`. Copy the needed pages into `images/` with space-free names.
- **Reuse, do not duplicate.** `clinical_background.tex` (sources, error argument) and `endpoints.tex` (endpoint and cohort construction). Citation keys already in `../general/references.bib`: `hammer1996`, `kaslow1987`, `delta1996`, `larder1989`, `concorde1994`, `hernan2010`.
- **Caveats.** The ATE numbers are from the provisional (v5) run. The heterogeneity and confounding prose is figure-driven. Specific subgroup and $c(x)$ values stay qualitative or placeholder until read off the final figures.